

Soorej Santhoshkumar Nair

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EDUCATION

University of Colorado, Boulder

Master of Science in Data Science — GPA: 4.00

Aug 2025 – May 2027

Boulder, CO

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering in Computer Science

Aug 2017 – May 2021

Pilani, India

1 Technical Skills

- **Skills:** Python, Java, C++, Rust, SQL, MATLAB, PowerBI, Tableau, Numpy, Pandas, Matplotlib, TensorFlow, Keras, Hugging Face, OpenCV, Scikit-learn, Linux, Databricks MLflow, Statistics, Diffusion Models, Vision-Language Models, GANs, Autoencoders, LangChain, Agentic AI, RAGs, Image Processing, Map-reduce, Reinforcement learning, Machine Learning, Regression, Classification, AWS, GCP, Docker, Kubernetes, Git, MLflow, Airflow, Google Cloud Platform, Hadoop, Spark

2 Summary

Highly motivated and detail-oriented data scientist with a strong background in machine learning, deep learning, and natural language processing. Proficient in a range of programming languages, including Python, Java, and C++. Experienced in working with large datasets and developing predictive models to drive business growth.

3 Experience

- **Senior Data Scientist**
XPO Logistics
 - Developed and deployed a Random Forest churn prediction model that achieved 90% precision in the top decile, preventing over \$240M in potential revenue loss.
 - Designed and implemented an OCR-based computer vision system that processed 60K+ bills of lading daily, extracting structured data into BigQuery and SQL-based pipelines.
 - Created an end-to-end LLM-based pipeline using Google Gemini, GCP, and Vertex AI, integrating CV, NLP, and OCR workflows for automated invoice extraction and classification.
- **Data Scientist**
XPO Logistics
 - Partnered with McKinsey to develop an ML-based logistics simulation model that improved freight handling efficiency by 54%.

- Developed a computer vision audit platform using GCP, Pub/Sub, SQL, and Vertex AI Pipelines, processing 80K+ documents daily and enhancing data quality metrics.
- Created CNN-based models for accessorial optimization and trailer image analysis, generating \$1.3M in incremental revenue.

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Associate Data Scientist

XPO Logistics

- Built decision tree models to detect fraudulent claims using Vertex AI and SQL queries, generating \$15K monthly.
- Developed ARIMA-based time series forecasting models in R, Python, and BQML to predict employee attrition, communicating insights through data visualization and storytelling.

4 Projects

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Multimodal Vision-Language Model (VLM) from Scratch

- Implemented a custom Vision-Language Model in PyTorch, building the core Transformer architecture and integrating a SigLIP-based vision encoder with a language model via a custom MLP projector.
- Optimized training and inference by implementing causal masking and RMSNorm, improving stability and enabling efficient token generation for 7-DoF robotic action sequences.

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Transformer-based Hindi Poetry Generator

- Built a custom decoder-only Transformer-based generative model leveraging natural language processing and statistical text analysis on data fetched from web scraping.
- Applied data wrangling and model evaluation metrics to assess creativity and linguistic coherence.

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Implementation of Sparsity Invariant CNNs

- Implemented sparse convolutional neural networks (SCNNs) for LiDAR data, applying statistical validation and quantitative performance measurement to compare against traditional CNNs.